Vivaan Bhandary

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Education

University of British Columbia — Vancouver, BC
Bachelor of Science in Computer Science — *Sep. 2021 – May 2026*
GPA: 4.33, Dean's Honour List: 2022, 2023, 2024. Dean's Scholar 2026, Graduated with Distinction

Experience

Junior Quality Assurance Engineer Co-op — Sept 2024 – Apr 2025
Trulioo — *Vancouver, BC*

- Progressed to Risk Team's **solo QA & QA Lead** within four months, managing end-to-end testing, deployments, and regression for critical risk monitoring systems
- **Optimized automated API test suites** for daily CI/CD pipelines, accelerating execution times by **4.5x**
- **Engineered comprehensive Postman collections** for **REST APIs** using dynamic variables and external data integrations to scale automated testing
- Developed and maintained mock API sandboxes to simulate third-party data sources and machine learning responses, enabling the dynamic testing of complex failure combinations and edge cases
- Led QA delivery for a high-priority data validation feature designed to intercept and quarantine false-positive data, managing test plans to ensure a zero-defect release under tight sprint deadlines
- Coordinated cross-functional testing strategies during a company-wide API standardization initiative, synchronizing automated test coverage across multiple distributed teams to ensure seamless integration

Artificial Intelligence Internship — July 2024 – Aug 2024
Safe Security — *Mumbai, India*

- Researched Retrieval-Augmented Generation (RAG) frameworks to optimize LLMs for accurate data retrieval
- Deployed a custom LLaMA architecture with RAG context to significantly improve context-aware AI responses

Internship — June. 2023 – Aug 2023
Infectious Advertising — *Mumbai, India*

- Overhauled the internal Job List Manager by transitioning it from a shared Google Doc to Excel, incorporating automation to generate daily work tasks and charts by analyzing employee data
- Enhanced user-friendliness by adding combo boxes and data validation to the new system to facilitate a smooth transition for employees while also being easy to adopt
- Actively engaged in brainstorming sessions for the development of promotional ideas and writing copy for advertising materials for various clients
- Edited and updated website content for a client to convey its information with more clarity and broader appeal using trendier terminology

Projects

Reloaded Dice - Video Game and Engine | *C++, Git, EnTT, Godot* — Jan 2026 – Apr 2026

- Directed a 6-person Agile team through 2-week sprints, integrating playtest feedback to refine core gameplay loops
- Architected a C++ game engine using the EnTT ECS framework for highly performant and scalable game logic
- Optimized core engine processes with the team by implementing spatial hashing and collision layers for efficient physics calculations, and resolved a major rendering bottleneck to reduce GPU overhead
- Designed and programmed a dynamic, synergistic upgrade system, managing complex entity interactions and state changes to scale gameplay mechanics
- Composed and integrated a dynamic music and sound system for the game, increasing player feedback for important actions and telegraphing gameplay elements

Graphics and Animation Projects | *JavaScript, GLSL, Python, speech_recognition* — Sept 2025 – Apr 2026

- Developed an interactive 3D graphics engine utilizing WebGL and custom GLSL shaders to render real-time environments
- Implemented multi-pass rendering for dynamic shadows and a multi-camera system to simulate movable, real-time portal effect

- Engineered an audio-driven facial animation system that uses state interpolation to smoothly transition between expressions based on real-time voice commands
- Developed a custom animation pipeline that supports keyframe saving and applies inverse kinematics (IK) interpolation to compute fluid, dynamic motion
- Wrote a physics-based cloth simulation engine featuring interactive collision detection, particle external forces, and custom cloth-tearing logic

Campus Information Query Engine | *TypeScript, Git, React, REST APIs* — Sept 2023 – Dec 2023

- Led Agile development of a TypeScript application, managing iterative sprints and weekly scrum meetings
- Parsed JSON and HTML campus data to create datasets for complex search and query operations
- Built secure RESTful API endpoints connecting the backend server with a React frontend
- Built a robust testing suite utilizing black-box, white-box, mutation, and unit testing

Data Analysis & Machine Learning Projects — Sept 2021 - Apr 2022

- Developed a K-Nearest Neighbors (KNN) regression model in R ($R^2 = 0.9$) to analyze and predict tennis player prize money based on ranking data
- Conducted hypothesis testing & generated bootstrap distributions using R to analyze regional demographics

Zone Chasing Game | *Java, Git, Java Swing* — Jan 2022 - Apr 2022

- Developed an object-oriented Java game utilizing the Swing library, featuring a dynamic GUI with custom gameplay animations
- Implemented JSON-based data persistence for save/load functionality and engineered a custom console logger to track object state changes
- Designed a comprehensive testing suite to streamline debugging processes and minimize gameplay defects

Generative Math Quizzer| *Java, MySQL* — Jan 2021

- Developed a Java application to dynamically generate and evaluate math assessments, utilizing flowchart prototyping to integrate iterative client feedback
- Architected a secure, password-protected RDBMS backend using MySQL to efficiently manage and store student test data

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C/C++, HTML, R, SQL, Erlang, PHP, Kotlin, GLSL
Frameworks & Libraries: React, Node.js, scikit-learn, Pandas, NumPy, Matplotlib, three.js, EnTT, NVIDIA warp
Software & Tools: Git, GitHub, Unity, Godot, Adobe Premiere Pro, VS Code, Visual Studio, PyCharm
Data Analysis & Visualization: Data Mining, Machine Learning, Data Visualization
Relevant Coursework: Video Game Programming, Computer Graphics & Animation, Machine Learning & Data Mining, Parallel Computation, Computer Networking, Computer Hardware & Operating Systems, Software Engineering & Construction, Algorithms and Data Structures, Data Science

Additional Certificates & Awards

1st Place at Trulioo's Fakeathon | *Company-wide Deepfake Detection* — Apr 2025
Wildest Idea Prize | *UBC Learning Analytics Hackathon* — Oct 2024
1st Place Prize | *UBC Game Design Competition* — Apr 2024